EXERCISE 2

Demagnetizing Field via FFT

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The demagnetizing (stray) field of a magnetization distribution $\vec{m}(\vec{r})$ is a convolution, $\vec{h}(\vec{r}) = \int \mathbf{K}(\vec{r} - \vec{r}')\,\vec{m}(\vec{r}')\,d\vec{r}'$. On a grid this costs $\mathcal{O}(N^2)$ directly, but only $\mathcal{O}(N \log N)$ with the FFT. Cells of unit size.

1 Demagnetizing field in 1D via FFT convolution. A row of N cells along the x-axis with $m_x = 1$ on a segment and 0 elsewhere; sources and evaluation points lie on the x-axis only ($y = z = 0$).

- **a)** The field of a point dipole $\vec{m}$ at the origin is $\vec{h}(\vec{r}) = \frac{1}{4\pi r^3}\left(3(\vec{m}\cdot\hat{r})\,\hat{r} - \vec{m}\right)$.
  Plot the field of a single dipole and derive the 1D kernel $h_x = K_{xx}(x)\,m_x$ on the x-axis.
  How is the self-interaction ($x = 0$) treated?
- **b)** Compute $h_x = K_{xx} * m_x$ by direct summation.
  What happens at the ends of the segment, and in the middle?
- **c)** Use the convolution theorem, $\hat{h}_k = \hat{K}_k\,\hat{m}_k$.
  How can we get a linear (instead of a circular) convolution?
  Why is zero padding to $2N$ necessary, and how must the kernel be arranged?
- **d)** Compare timings of direct and FFT convolution for different system sizes N.
  What is the asymptotic scaling of both algorithms?
  What happens for $N \neq 2^m$?
- **e)** Try to optimize the FFT algorithm further and benchmark your improvements.
  How can the algorithm be further optimized?

2 Extend to 3D. Same dipole kernel as in 1a), now with $\vec{m}$ and $\vec{h}$ on an $n \times n \times n$ grid.

- **a)** Write the dipole field as $\vec{h} = \mathbf{K}(\vec{r})\,\vec{m}$ with a 3×3 tensor $\mathbf{K}$ and implement $\vec{h} = \mathbf{K} * \vec{m}$ with the FFT.
  How many independent kernel components and how many FFTs are needed?
- **b)** Validate that the calculated results are correct.
  How can you check for correctness?
  How large is the error, and what does it depend on?
- **c)** Compute and visualize $\vec{h}$ of a uniformly magnetized cube and sphere
  Why is the field uniform inside the sphere, but not inside the cube?
- **d)** Compare with the magnetic charge formulation: $\rho = -\nabla\cdot\vec{m}$, $\phi = K_c * \rho$ with $K_c(r) = 1/(4\pi r)$, $\vec{h} = -\nabla\phi$.
  Compare results, accuracy near the surface and computational cost.
  Why are both formulations equivalent?



